C 2-3

1.(1) $\int_{-\infty}^{+\infty}\psi_m^*\psi_n\,dx=\int_0^a\sqrt{\frac{2}{a}}\cdot\sin\left(\frac{m\pi}{a}x\right)\cdot\sqrt{\frac{2}{a}}\sin\left(\frac{n\pi}{a}x\right)dx$

$=\frac{2}{a}\cdot\int_0^a\sin\left(\frac{m\pi}{a}x\right)\sin\left(\frac{n\pi}{a}x\right)dx$

$=\frac{1}{a}\int_0^a\left[\cos\left(\frac{m-n}{a}\pi x\right)-\cos\left(\frac{m+n}{a}\pi x\right)\right]dx$

若$m\neq n$，上式 $=\left[\frac{1}{(m-n)\pi}\sin\left(\frac{m-n}{a}\pi x\right)-\frac{1}{(m+n)\pi}\sin\left(\frac{m+n}{a}\pi x\right)\right]\Big|_0^a$

$=\frac{1}{\pi}\cdot\left[\frac{\sin(m-n)\pi}{m-n}-\frac{\sin(m+n)\pi}{m+n}\right]$

$=0$

若$m=n$，则$\int_{-\infty}^{+\infty}\psi_m^*\psi_n\,dx=\frac{1}{a}\int_0^a\left(1-\cos\frac{2n}{a}\pi x\right)dx$

$=\frac{1}{a}x\Big|_0^a-\left(\sin\frac{2n}{a}\pi x\right)\Big|_0^a\cdot\frac{1}{a}\cdot\frac{a}{2n\pi}$

$=1-0=1$

则$\int_{-\infty}^{+\infty}\psi_m^*\psi_n\,dx=\begin{cases}0, & m\neq n\\ 1, & m=n\end{cases}$

$=\delta_{mn}$，得证.

1.(2) $\psi(x,0)=A\sin^3(\pi x/a) \quad (0\le x\le a)$

由于 $\sin 3\theta = 3\sin\theta - 4\sin^3\theta$，则 $\sin^3\theta = \frac{3}{4}\sin\theta - \frac{1}{4}\sin 3\theta$

$$\psi(x,0)=A\left[\frac{3}{4}\sin\left(\frac{\pi}{a}x\right)-\frac{1}{4}\sin\left(\frac{3\pi}{a}x\right)\right]$$

$\int_0^a \psi^2(x,0)\,dx=1$，则 $\int_0^a A^2\left[\frac{9}{16}\sin^2\left(\frac{\pi}{a}x\right)+\frac{1}{16}\sin^2\left(\frac{3\pi}{a}x\right)-\frac{6}{16}\sin\left(\frac{\pi}{a}x\right)\sin\left(\frac{3\pi}{a}x\right)\right]dx$

$$=\int_0^a A^2\left[\frac{9}{16}\sin^2\left(\frac{\pi}{a}x\right)+\frac{1}{16}\sin^2\left(\frac{3\pi}{a}x\right)\right]dx$$

$$=A^2\cdot\frac{9}{16}\cdot\frac{a}{2}+A^2\cdot\frac{1}{16}\cdot\frac{a}{2}=\frac{5}{16}A^2\cdot a=1$$

由 $\frac{5}{16}A^2\cdot a=1$ 解得 $A=\sqrt{\frac{16}{5a}}$

则 $\psi(x,0)=\sqrt{\frac{16}{5a}}\cdot\left[\frac{3}{4}\sin\left(\frac{\pi}{a}x\right)-\frac{1}{4}\sin\left(\frac{3\pi}{a}x\right)\right]$

$$=\sqrt{\frac{9}{5a}}\sin\left(\frac{\pi}{a}x\right)-\sqrt{\frac{1}{5a}}\sin\left(\frac{3\pi}{a}x\right)=\frac{3}{\sqrt{10}}\psi_1(x,0)-\frac{1}{\sqrt{10}}\psi_3(x,0)$$

则 $\psi(x,t)=\frac{3}{\sqrt{10}}\psi_1(x)\cdot e^{-i\frac{E_1}{\hbar}t}-\frac{1}{\sqrt{10}}\psi_3(x)\cdot e^{-i\frac{E_3}{\hbar}t}$

其中 $E_1=\frac{\pi^2\hbar^2}{2ma^2}$，$E_3=\frac{9\pi^2\hbar^2}{2ma^2}$

则 $\langle E\rangle=\frac{9}{10}\times E_1+\frac{1}{10}\times E_3=\frac{9\pi^2\hbar^2}{10\cdot ma^2}$

$\langle x\rangle=\int_0^a \psi^2(x)\,x\,dx=\int_0^a\left[\frac{9}{5a}\sin^2\left(\frac{\pi}{a}x\right)\cdot x+\frac{1}{5a}\sin^2\left(\frac{3\pi}{a}x\right)\cdot x-\frac{6}{5a}\sin\left(\frac{\pi}{a}x\right)\sin\left(\frac{3\pi}{a}x\right)\cdot x\right]dx$

$\int_0^a\frac{9}{5a}\sin^2\left(\frac{\pi}{a}x\right)\cdot x\,dx=\int_0^a\frac{9}{10}\left(1-\cos\frac{2\pi}{a}x\right)\cdot x\,dx\cdot\frac{1}{a}$

$$=\left\{\frac{9}{10}\cdot\frac{1}{2}x^2-\frac{9}{10}\left(\frac{a}{2\pi}\right)^2\cdot\left[\frac{2\pi}{a}\cdot x\sin\left(\frac{2\pi}{a}x\right)+\cos\left(\frac{2\pi}{a}x\right)\right]\right\}\Bigg|_0^a\times\frac{1}{a}$$

$$=\frac{9}{20}a$$

$\int_0^a\frac{1}{5}\sin^2\left(\frac{3\pi}{a}x\right)x\,dx=\frac{1}{20}a$，$\int_0^a\frac{6}{5a}\sin\left(\frac{\pi}{a}x\right)\sin\left(\frac{3\pi}{a}x\right)\cdot x\,dx=0$.

故 $\langle x\rangle=\frac{9}{20}a+\frac{1}{20}a=\frac{a}{2}$.

综上，$A=\sqrt{\frac{16}{5a}}$，$\langle x\rangle=\frac{a}{2}$，$\langle E\rangle=\frac{9\pi^2\hbar^2}{10ma^2}$

$\psi(x,t)=\frac{3}{\sqrt{10}}\psi_1(x)e^{-i\frac{E_1}{\hbar}t}-\frac{1}{\sqrt{10}}\psi_3(x)e^{-i\frac{E_3}{\hbar}t}$，$E_n=\frac{\pi^2\hbar^2}{2ma^2}\cdot n^2$.

2.① 证明 $\langle\psi|\phi\rangle=\langle\phi|\psi\rangle^*$

$$|\psi\rangle=\begin{bmatrix}\psi_1\\ \psi_2\\ \vdots\\ \psi_n\end{bmatrix}\quad |\phi\rangle=\begin{bmatrix}\phi_1\\ \phi_2\\ \vdots\\ \phi_n\end{bmatrix}$$，则 $\langle\psi|=[\psi_1^*\ \psi_2^*\ \psi_3^*\ \dots\ \psi_n^*]$

$\langle\phi|=[\phi_1^*\ \phi_2^*\ \phi_3^*\ \dots\ \phi_n^*]$

$$左式=\langle\psi|\phi\rangle=[\psi_1^*\ \psi_2^*\ \dots\ \psi_n^*]\cdot\begin{bmatrix}\phi_1\\ \phi_2\\ \vdots\\ \phi_n\end{bmatrix}$$

$$=\psi_1^*\phi_1+\psi_2^*\phi_2+\psi_3^*\phi_3+\dots+\psi_n^*\phi_n.$$

$$\langle\phi|\psi\rangle=[\phi_1^*\ \phi_2^*\ \dots\ \phi_n^*]\begin{bmatrix}\psi_1\\ \psi_2\\ \vdots\\ \psi_n\end{bmatrix}=\phi_1^*\psi_1+\phi_2^*\psi_2+\dots+\phi_n^*\psi_n$$

则右式 $=\langle\phi|\psi\rangle^*=\psi_1^*\phi_1+\psi_2^*\phi_2+\psi_3^*\phi_3+\dots+\psi_n^*\phi_n=$ 左式 得证.

② 证明 $\langle\psi|A|\phi\rangle=\langle\phi|A^\dagger|\psi\rangle^*$ $\quad \underset{1\times n\ \ n\times n\ \ n\times 1}{\langle\psi|A|\phi\rangle}$

左式 $=\langle\psi|A|\phi\rangle=\langle\psi|A\phi\rangle=\langle A\phi|\psi\rangle^*$

则证明 $\langle\phi|A^\dagger=\langle A\phi|$，即可证明 $\langle\psi|A\phi\rangle=\langle\phi|A^\dagger|\psi\rangle^*$

$$|\phi\rangle=\begin{bmatrix}\phi_1\\ \phi_2\\ \vdots\\ \phi_n\end{bmatrix},\quad A=\begin{bmatrix}a_{11}&a_{12}&\dots&a_{1n}\\ a_{21}&a_{22}&\dots&a_{2n}\\ \vdots&\vdots&\ddots&\vdots\\ a_{n1}&a_{n2}&\dots&a_{nn}\end{bmatrix}.$$

$$\langle\phi|A^\dagger=[\phi_1^*\phi_2^*\dots\phi_n^*]\cdot\begin{bmatrix}a_{11}^*&a_{21}^*&\dots&a_{n1}^*\\ a_{12}^*&a_{22}^*&&a_{n2}^*\\ \vdots&&\ddots&\\ a_{1n}^*&a_{2n}^*&\dots&a_{nn}^*\end{bmatrix}.$$

= … 见后一页

$$=[\phi_1^* a_{11}^* + \phi_2^* a_{12}^* + \phi_3^* a_{13}^* + \dots + \phi_n^* a_{1n}^*, \phi_1^* a_{21}^* + \phi_2^* a_{22}^* + \dots + \phi_n^* a_{2n}^*, \dots, \phi_1^* a_{n1}^* + \phi_2 a_{n2}^* + \dots + \phi_n^* a_{nn}^*]$$

$$\langle A\phi| = |A\phi\rangle^\dagger$$

$$= \left(\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix} \cdot \begin{bmatrix} \phi_1 \\ \phi_2 \\ \vdots \\ \phi_n \end{bmatrix} \right)^\dagger = \begin{bmatrix} a_{11}\phi_1 + a_{12}\phi_2 + \dots + a_{1n}\phi_n \\ a_{21}\phi_1 + a_{22}\phi_2 + \dots + a_{2n}\phi_n \\ \vdots \\ a_{n1}\phi_1 + a_{n2}\phi_2 + \dots + a_{nn}\phi_n \end{bmatrix}^\dagger$$

$$= [a_{11}^*\phi_1^* + a_{12}^*\phi_2^* + \dots + a_{1n}^*\phi_n^*, a_{21}^*\phi_1^* + a_{22}^*\phi_2^* + \dots + a_{2n}^*\phi_n^*, a_{n1}^*\phi_1^* + a_{n2}^*\phi_2^* + \dots + a_{nn}^*\phi_n^*]$$

$$= \langle\phi|A^\dagger$$

故 $\langle\phi|A^\dagger|\psi\rangle^* = \langle A\phi|\psi\rangle^* = \langle\psi|A\phi\rangle = \langle\psi|A|\phi\rangle$，得证

或者根据 $(AB)^\dagger = B^\dagger A^\dagger$ 可证 $\langle\phi|A^\dagger = \langle A\phi|$

因为 $\langle A\phi| = (|A\phi\rangle)^\dagger = (A|\phi\rangle)^\dagger = \langle\phi|A^\dagger$，得证 $\langle\phi|A^\dagger| = \langle A\phi|$

则 $\langle\phi|A^\dagger|\psi\rangle^* = \langle\psi|A|\phi\rangle$ 得证

$\cancel{3.\ \frac{d}{dt}\int_{-\infty}^{+\infty}\psi_1^*\psi_2\,dx}$

$\cancel{=\frac{d}{dt}\int_{-\infty}^{+\infty}\psi_1^*(x)e^{i\frac{E_1}{\hbar}t}\cdot\psi_2(x)e^{-i\frac{E_2}{\hbar}t}\,dx}$

3. 证明 $\frac{d}{dt}\int_{-\infty}^{+\infty}\psi_1^*\psi_2\,dx=0$

$$\frac{d}{dt}\int_{-\infty}^{+\infty}\psi_1^*\psi_2\,dx=\int_{-\infty}^{+\infty}\left(\psi_1^*\frac{\partial\psi_2}{\partial t}+\frac{\partial\psi_1^*}{\partial t}\cdot\psi_2\right)\cdot dx.$$

含时薛定谔方程：$i\hbar\frac{\partial\psi}{\partial t}=-\frac{\hbar^2}{2m}\cdot\frac{\partial^2\psi}{\partial x^2}+V\psi$ ①

$\downarrow$

取共轭：$-i\hbar\frac{\partial\psi^*}{\partial t}=-\frac{\hbar^2}{2m}\cdot\frac{\partial^2\psi^*}{\partial x^2}+V\psi^*$ ②

则 $\frac{\partial\psi_2}{\partial t}$ 由①式可得：$\frac{\partial\psi_2}{\partial t}=-\frac{\hbar}{2mi}\cdot\frac{\partial^2\psi_2}{\partial x^2}+\frac{V\psi_2}{i\hbar}$

$\frac{\partial\psi_1^*}{\partial t}$ 由②式可得：$\frac{\partial\psi_1^*}{\partial t}=\frac{\hbar}{2mi}\cdot\frac{\partial^2\psi_1^*}{\partial x^2}-\frac{V\psi_1^*}{i\hbar}$

代回 $\frac{d}{dt}\int_{-\infty}^{+\infty}\psi_1^*\psi_2\,dx=\int_{-\infty}^{+\infty}\left(\psi_1^*\cdot\frac{\partial\psi_2}{\partial t}+\frac{\partial\psi_1^*}{\partial t}\cdot\psi_2\right)dx$

$$=\int_{-\infty}^{+\infty}\left[\psi_1^*\cdot\left(-\frac{\hbar}{2mi}\cdot\frac{\partial^2\psi_2}{\partial x^2}+V\frac{\psi_2}{i\hbar}\right)+\left(\frac{\hbar}{2mi}\frac{\partial^2\psi_1^*}{\partial x^2}-V\frac{\psi_1^*}{i\hbar}\right)\psi_2\right]dx$$

$$=\int_{-\infty}^{+\infty}\left[\psi_1^*\frac{\partial^2\psi_2}{\partial x^2}\cdot\frac{-\hbar}{2mi}+\frac{\hbar}{2mi}\cdot\psi_2\cdot\frac{\partial^2\psi_1^*}{\partial x^2}\right]dx \quad ③$$

对于：$\int_{-\infty}^{+\infty}\psi_1^*\frac{\partial^2\psi_2}{\partial x^2}dx$，用分部积分处理.

$$\text{上式}=\int_{-\infty}^{+\infty}\psi_1^*\cdot d\left(\frac{\partial\psi_2}{\partial x}\right)=\psi_1^*\frac{\partial\psi_2}{\partial x}\Big|_{-\infty}^{+\infty}-\int_{-\infty}^{+\infty}\frac{\partial\psi_2}{\partial x}\cdot d(\psi_1^*)$$

$$= \psi_1^* \frac{\partial \psi_2}{\partial x}\Big|_{-\infty}^{+\infty} - \int_{-\infty}^{+\infty} \frac{\partial \psi_2}{\partial x} \cdot \frac{\partial \psi_1^*}{\partial x} dx$$

由于波函数在无穷远处为0，则上式 $= -\int_{-\infty}^{+\infty} \frac{\partial \psi_2}{\partial x} \cdot \frac{\partial \psi_1^*}{\partial x} dx$. ④

同理得 $\int_{-\infty}^{+\infty} \psi_2 \cdot \frac{\partial^2 \psi_1^*}{\partial x^2} dx = \int_{-\infty}^{+\infty} \psi_2 \cdot d\left(\frac{\partial \psi_1^*}{\partial x}\right)$

$$= \psi_2 \cdot \frac{\partial \psi_1^*}{\partial x}\Big|_{-\infty}^{+\infty} - \int_{-\infty}^{+\infty} \frac{\partial \psi_1^*}{\partial x} \cdot \frac{\partial \psi_2}{\partial x} \cdot dx .$$

$$= -\int_{-\infty}^{+\infty} \frac{\partial \psi_2}{\partial x} \cdot \frac{\partial \psi_1^*}{\partial x} \cdot dx . \quad ⑤$$

将④、⑤代入③中，可得：

$$\frac{d}{dt} \int_{-\infty}^{+\infty} \psi_1^* \psi_2 dx = -\frac{\hbar}{2mi} \cdot \left(-\int_{-\infty}^{+\infty} \frac{\partial \psi_2}{\partial x} \cdot \frac{\partial \psi_1^*}{\partial x} \cdot dx\right) + \frac{\hbar}{2mi} \cdot \left(-\int_{-\infty}^{+\infty} \frac{\partial \psi_2}{\partial x} \cdot \frac{\partial \psi_1^*}{\partial x} dx\right)$$

$$= 0 .$$

则原命题得证.